Michael A Reefe

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Michael-Reefe
Michael Reefe
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Key Research Interests

- Galaxy clusters, cooling flows, the intracluster/circumgalactic medium
- AGN feeding & feedback, supermassive black hole & host galaxy evolution
- Coronal emission lines, multiwavelength astronomy, integral field spectroscopy

Education

PhD, Physics, Massachusetts Institute of Technology — 2022 – Present
Thesis: *The Circumgalactic Medium of Brightest Cluster Galaxies*
Advisor: Prof. Michael McDonald

BS, Physics (Astrophysics concentration), George Mason University — 2018 – 2022
Summa cum laude
Honors Thesis: *CLASS: Coronal Line Activity Spectroscopic Survey*
Advisor: Prof. Shobita Satyapal

Fellowships, Honors, and Awards

NSF, Graduate Research Fellowship — $159k — 2022 – 2027
Awarded annually to ~ 2500 US undergraduate/graduate students pursuing STEM degrees.

MIT Physics, Pauline Whiteman Fellowship — $95k — 2022 – 2023
Full graduate tuition and stipend for 1 year of study.

GMU COS, Dean's Award for Excellence in Academics/Research — $1k — 2022
College-wide (College of Science) award to a single undergraduate student for outstanding performance in both academics and research.

GMU COS, Graduation Student Speaker — 2022
Nominated and selected as the student speaker for the College of Science's graduation ceremony, out of the undergraduate and graduate student bodies.

GMU Physics, Outstanding Undergraduate Research Award — 2022

GMU Physics, Outstanding Graduating Senior Award — 2022

GMU, Mason Distinction Scholarship — $8k — 2018 – 2022

GMU Physics, Outstanding Learning Assistant Award — $150 — 2021

GMU Physics, Osher Lifelong Learning Institute Scholarship — $500 — 2020

Publications

Listings: ADS, arXiv, Google Scholar | Citations: 627, h-index: 13 | 1 publication in *Nature*

First Author

10. **M. Reefe**, M. McDonald, M. Chatzikos et al., "O VI Emission in Starbursting BCGs: An HST-COS Extension of "Reigniting the FUSE"." *arXiv e-prints* arXiv:2610.34579 (2026), submitted to ApJ.

9. **M. Reefe**, M. McDonald, G. L. Bryan et al., "Reigniting the FUSE II: O VI Emission in Massive Elliptical Galaxies." *arXiv e-prints* arXiv:2609.13368 (2026), submitted to ApJ.

8. **M. Reefe**, M. McDonald, E. Miller et al., "Reigniting the FUSE I: Star Formation in Massive Elliptical Galaxies." *arXiv e-prints* arXiv:2609.05675 (2026), submitted to ApJ.

7. **M. Reefe**, M. McDonald, M. Chatzikos et al., "Cold Gas and Star Formation in the Phoenix Cluster with JWST." *ApJ* **989** (2), 156 (2025).

6. **M. Reefe**, M. McDonald, M. Chatzikos et al., "Directly imaging the cooling flow in the Phoenix cluster." *Nature* **638** (8050), 360–364 (2025).

5. **M. Reefe**, S. Satyapal, R. O. Sexton et al., "Nuclear Activity in the Low-metallicity Dwarf Galaxy SDSS J0944-0038 : A Glimpse into the Primordial Universe." *ApJL* **946** (2), L38 (2023).

4. **M. Reefe**, R. O. Sexton, S. M. Doan et al., "CLASS Survey Description: Coronal-line Needles in the SDSS Haystack." *ApJS* **265** (1), 21 (2023).

3. **M. Reefe**, S. Satyapal, R. O. Sexton et al., "CLASS: Coronal Line Activity Spectroscopic Survey." *ApJ* **936** (2), 140 (2022).

2. **M. Reefe**, R. Luque, E. Gaidos et al., "A Close-in Puffy Neptune with Hidden Friends: The Enigma of TOI 620." *AJ* **163** (6), 269 (2022).

1. **M. Reefe**, O. Alfaro, S. Foster et al., "Asynchronous object-oriented approach to the automation of the 0.8-meter George Mason University campus telescope in Python." *JATIS* **8**, 027002 (2022).

CO-AUTHOR .....

22. J. Lillo-Box, C. Cifuentes, O. Balsalobre-Ruza et al. incl. **M. Reefe**, "CAFE follow-up of TESS hot Jupiter candidates left behind: I. Five newly confirmed planets and a false positive." *A&A* **713**, A272 (2026).

21. M. Marquis, J. Hlavacek-Larrondo, O. Pereira et al. incl. **M. Reefe**, "Mapping gas accretion and stellar kinematics to sub-kiloparsec scales in NGC 4696 with JWST/NIRSpec." *arXiv e-prints* arXiv:2608.07732 (2026), submitted to ApJ.

20. J. Hlavacek-Larrondo, H. Choi, M. Guo et al. incl. **M. Reefe**, "JWST Reveals How Black Holes are Fed: Kiloparsec-scale Multiphase Filaments Feed Subkiloparsec Circumnuclear Disks." *ApJL* **1006** (1), L9 (2026).

19. V. Cammelli, M. Gaspari, F. Barbani et al. incl. **M. Reefe**, "BlackHoleWeather – Jet-regulated chaotic cold accretion across the meso scale: Variability and kinematics." *arXiv e-prints* arXiv:2605.27511 (2026), submitted to A&A.

18. V. Cammelli, M. Gaspari, O. Piana et al. incl. **M. Reefe**, "BlackHoleWeather – Jet-regulated chaotic cold accretion across the meso scale: Morphology and thermodynamics." *arXiv e-prints* arXiv:2605.27503 (2026), submitted to A&A.

17. M. McDonald, G. Khullar, D. Lagattuta et al. incl. **M. Reefe**, "A Kiloparsec-scale Stellar Cavity in the Center of A402-BCG May Be Caused by Dynamic Interactions with an Ultramassive Black Hole." *ApJL* **1002** (1), L19 (2026).

16. S. Doan, S. Satyapal, **M. Reefe** et al., "The CLASS Quasar Catalog: Coronal Line Activity in Type 1 SDSS Quasars." *ApJS* **280** (2), 57 (2025).

15. W. Matzko, S. Satyapal, **M. Reefe** et al., "An Increase in the Prevalence of Ionized Outflows in Galaxies with Coronal Line Emission: Feedback in Action?" *ApJ* **984** (2), 170 (2025).

14. C. R. Mann, P. A. Dalba, D. Lafrenière et al. incl. **M. Reefe**, "Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. III. Recovery and Confirmation of a Temperate, Mildly Eccentric, Single-transit Jupiter Orbiting TOI-2010." *AJ* **166** (6), 239 (2023).

13. J. M. Wittrock, P. P. Plavchan, B. L. Cale et al. incl. **M. Reefe**, "Validating AU Microscopii d with Transit Timing Variations." *AJ* **166** (6), 232 (2023).

12. J. E. Rodriguez, S. N. Quinn, A. Vanderburg et al. incl. **M. Reefe**, "Another shipment of six short-period giant planets from TESS." *MNRAS* **521** (2), 2765–2785 (2023).

11. R. W. Pfeifle, B. Rothberg, K. A. Weaver et al. incl. **M. Reefe**, "The Messy Nature of Fiber Spectra: Star-Quasar Pairs Masquerading as Dual Type 1 AGNs." *ApJ* **945** (2), 167 (2023).

10. S. W. Yee, J. N. Winn, J. D. Hartman et al. incl. **M. Reefe**, "The TESS Grand Unified Hot Jupiter Survey. II. Twenty New Giant Planets." *ApJS* **265** (1), 1 (2023).

9. R. W. Pfeifle, S. Satyapal, C. Ricci et al. incl. **M. Reefe**, "NuSTAR Observes Two Bulgeless Galaxies: No Hard X-Ray AGN Detected in NGC 4178 or J0851+3926." *ApJ* **943** (2), 109 (2023).

8. M. El Mufti, P. P. Plavchan, H. Isaacson et al. incl. **M. Reefe**, "TOI 560: Two Transiting Planets Orbiting a K Dwarf Validated with iSHELL, PFS, and HIRES RVs." *AJ* **165** (1), 10 (2023).

7. J. M. Wittrock, S. Dreizler, **M. Reefe** et al., "Transit Timing Variations for AU Microscopii b and c." *AJ* **164** (1), 27 (2022).

6. E. A. Gilbert, T. Barclay, E. V. Quintana et al. incl. **M. Reefe**, "Flares, Rotation, and Planets of the AU Mic System from TESS Observations." *AJ* **163** (4), 147 (2022).

5. B. L. Cale, **M. Reefe**, P. Plavchan et al., "Diving Beneath the Sea of Stellar Activity: Chromatic Radial Velocities of the Young AU Mic Planetary System." *AJ* **162** (6), 295 (2021).

4. A. Fukui, J. Korth, J. H. Livingston et al. incl. **M. Reefe**, "TOI-1749: an M dwarf with a Trio of Planets including a Near-resonant Pair." *AJ* **162** (4), 167 (2021).

3. A. Osborn, D. J. Armstrong, B. Cale et al. incl. **M. Reefe**, "TOI-431/HIP 26013: a super-Earth and a sub-Neptune transiting a bright, early K dwarf, with a third RV planet." *MNRAS* **507** (2), 2782–2803 (2021).

2. J. Teske, S. X. Wang, A. Wolfgang et al. incl. **M. Reefe**, "The Magellan-TESS Survey. I. Survey Description and Midsurvey Results." *ApJS* **256** (2), 33 (2021).

1. S. Dreizler, I. J. M. Crossfield, D. Kossakowski et al. incl. **M. Reefe**, "The CARMENES search for exoplanets around M dwarfs. LP 714-47 b (TOI 442.01): populating the Neptune desert." *A&A* **644**, A127 (2020).

PRESENTATIONS

(* = upcoming)

INVITED TALKS (COLLOQUIA & SEMINARS)

Reigniting the FUSE: A Modern Reanalysis of the Cooling Flow Problem with Archival UV Spectroscopy *Dec 2026
Galaxy Cluster Seminar, Center for Astrophysics | Harvard & Smithsonian

Coronal Gas in the circumgalactic medium of brightest cluster galaxies *Nov 2026
Special Seminar, New Mexico State University

Reigniting the FUSE: A Modern Reanalysis of the Cooling Flow Problem with Archival UV Spectroscopy *Nov 2026
Seminar, NASA Goddard Space Flight Center

Coronal Gas in the circumgalactic medium of brightest cluster galaxies *Nov 2026
Physics Colloquium, George Mason University

Reigniting the FUSE II: O VI Emission in Massive Elliptical Galaxies Sep 2026
Galread Seminar Series, Princeton University

Fully Tracing Cooling Gas in the Phoenix Cluster with JWST Mar 2025
Journal Club Series, MIT Kavli Institute

AGN feedback in an infant galaxy cluster: the LOFAR-Chandra view of the giant FRII radio galaxy J103025+052430 at z=1.7 Mar 2023
Journal Club Series, MIT Kavli Institute

CONTRIBUTED TALKS & POSTERS

Reigniting the FUSE: A Reanalysis of the Cooling Flow Problem with Archival UV Spectroscopy *Dec 2026
[Talk] BlackHoleWeather II, Sexten, IT

Coronal Gas in the circumgalactic medium of brightest cluster galaxies Dec 2025
[Talk] BlackHoleWeather I, Sexten, IT

Directly Observing the Cooling Flow in the Phoenix Cluster with JWST Oct 2025
[Talk] Multi-phase, Multi-temperature, and Complex, Olbia, IT

Directly Observing the Cooling Flow in the Phoenix Cluster with JWST Jul 2025
[Poster] Tracing Cosmic Evolution with Galaxy Clusters V, Sexten, IT

Mapping the Cooling Flow in the Phoenix Cluster with JWST and Chandra Dec 2024
[Talk] 25 Years of Science with Chandra, Boston, MA, US

Shaken or stirred? Dynamics of the coronal temperature gas in the Phoenix Cluster Jan 2024
[Talk] 243rd AAS Meeting, New Orleans, LA, US

A Large Scale Survey of Galaxies with Coronal Line Emission Selected from the Sloan Digital Sky Survey Jun 2022
[Poster] 240th AAS Meeting, Pasadena, CA, US

A Flexible Python Observatory Automation Framework for the George Mason University Campus Telescope Aug 2021
[Poster] TESS Science Conference II, Virtual

Automation of TESS Follow-up Observations with the GMU Campus Telescope Apr 2021
[Poster] GMU College of Science Undergraduate Research Colloquium, Virtual

An Asynchronous Object-Oriented Approach to Automation of the 0.8-meter George Mason University Campus Telescope in Python Jan 2021
[Poster] 237th AAS Meeting, Virtual

Conference Attendance

241st AAS Meeting, Seattle, WA, US Jan 2023

Teaching & Mentorship

Mentor, Physics Graduate Application Assistance Program, MIT 2024 – Present
PhysGAAP guides prospective PhD students through the MIT Physics application process, specifically targeting under-served communities and under-represented groups in STEM. I have mentored 5 prospective students through this program.

Research Project Leader, Warrior-Scholar Project, MIT Summer 2026
Led a group of 4 veteran students through an independent research project on numerical integration in Python, culminating in the simulation of a double pendulum system.

Mentor, Spectrum Peer Mentorship Program, GMU 2020 – 2022
Provided tutoring on academics and professional development to fellow undergraduate physics students at GMU.

Mentor, Aspiring Scientists' Summer Internship Program, GMU 2020 – 2021

Learning Assistant, PHYS 260: Electricity & Magnetism, GMU Fall 2019

Service

Member, Graduates Advising Graduate Admissions Committee, MIT 2024 – Present
Co-chair 2025–2026
GAGA advises the MIT Chair of Graduate Admissions from the graduate student perspective, aiming for increased diversity, equity, and inclusion, and organizes the Physics Graduate Application Assistance Program (see above).

Member, Physics Graduate Student Council, MIT 2023 – 2025

Vice President of Academic Advocacy 2024 – 2025
Webmaster 2023 – 2025
Worked as the primary graduate student advocate, leading initiatives to improve the quality of education, mentorship, and inclusiveness in department policy. E.g. creating guidelines for academic advisors, standardizing expectations for the oral qualifying exam, implementing professional development requirements, etc.

Organizer, Kavli Institute Graduate Student Lunch Series, MIT 2023 – 2024

Volunteer, Graduate School Fair, 241st AAS Meeting 2023

Faculty Search Undergraduate Liaison, Dept. of Physics & Astronomy, GMU 2022

Member, Spectrum, GMU 2020 – 2022

President 2021 – 2022
A student-led organization that works directly with the College of Science faculty to promote the enhancement of historically minoritized groups in STEM. Also runs a peer mentorship program (see above).

Scorer, MathCounts Competition, GMU 2020

Professional Collaborations

Member, BlackHoleWeather (PI: Gaspari) 2025 – Present

Member, JWST Galaxy Cluster Science Collaboration (PIs: Donahue, Guillard, Salomé) 2025 – Present

Outreach & Press

Member, Sidewalk Astrogazers, MIT 2023 – Present

- Many public sidewalk observing nights (2023 – Present)
- Booths and solar observing at the Cambridge Science Festival (2023, 2024)
- Trivia at the MIT Museum's *After Dark* series (2024)
- Exhibits during Massachusetts Space Week (2025)

Public Talk, Astronomy on Tap, Aeronaut Brewing Co, Somerville, MA 2025

Galaxy Clusters: From the subatomic to the extragalactic

Science Advisor, MIT Museum Tactile Cart Demonstration 2025

Provided scientific context for a tactile display of Stephan's Quintet, used in an interactive cart program for blind and visually impaired visitors to the museum.

Press Coverage, "Directly imaging the cooling flow in the Phoenix Cluster", Reefe et al., *Nature* 2025

Covered by NASA, CXO, MIT News, APS Physics Magazine, Space.com, Universe Today, and more

Panelist, ASSIP Career Day Graduate School Panel, GMU 2022

Panelist, Office of Fellowships NSF GRFP Cohort Workshop, GMU 2022

Software Products

Core Projects

Loki.jl, Lead developer 2022 – Present

A Julia package for modeling IFU spectroscopy in the optical, IR, and UV, designed with JWST data in mind. Now used by most of the community studying BCGs with JWST. Includes 11 example notebooks.

FSPS and Python-FSPS, Contributor 2025 – Present

A forked version of Charlie Conroy's Flexible Stellar Population Synthesis, and its corresponding Python bindings, that includes higher resolution stellar template libraries.

BADASS, Contributor 2022

A Python package that fits SDSS optical spectra of AGN using Bayesian methods. I wrote the IFU module designed to fit cubes from MaNGA and MUSE.

bifrost, Lead developer 2021 – 2022

A Python package for quickly searching millions of spectra for coronal line emission using both heuristics and a neural network.

-OmegaLambda, Lead developer 2020 – 2022

A Python package that fully automates the nightly observing operations of GMU's 0.8 m campus telescope and has facilitated hundreds of nights of automated observing.

Personal Projects and Ports

csplayer, Lead developer 2026 – Present

A desktop music player app, written in C, that displays the audio spectrum (Fourier transform) in real time.

AstroLib.jl, Contributor 2025

A general-purpose Julia library for standard astronomical routines, based on the Goddard IDL Astronomy User's Library. I ported a few functions for shifting and rebinning images.

VoronoiBinning.jl, Lead developer 2024

A Julia port of the `vorbin` Python package by Michele Cappellari.

SpectralResampling.jl, Lead developer 2024

A Julia port of the `SpectRes` Python package by Andrew Carnall.

Software Skills & Experience

Programming

Python, fluent with Python and its astronomy ecosystem, including but not limited to: numpy, scipy, astropy, matplotlib, pandas, numba, emcee, dynesty, prospector, etc. 6 years

Git/GitHub, highly experienced with collaborative workflows on GitHub 6 years

Bash/Shell, highly experienced in unix terminals and submitting HPCC jobs with SLURM 5 years

Julia, highly experienced in Julia's astronomy ecosystem, including the JuliaAstro project 4 years

IDL, basic experience running and translating programs from IDL to Python 2 years

Fortran, somewhat experienced, having worked with FSPS and its Python bindings 1 year

C, Make/CMake, basic experience, self-taught with Project Euler and csplayer 1 year

HTML/CSS/JS, enough experience to make a basic website 1 year

Analysis Programs: Cloudy, FSPS, CIAO, Xspec, Sherpa, DS9, AstroImageJ
Presentation: LaTeX, Matplotlib, Plotly, Microsoft Office Suite, Google Docs Editors
OS: macOS, Linux, Unix, Windows

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